

Ohm's Law & Current Problems

Homework Sheet

Ohm's Law

Problem 1

What happens to the current if the resistance in a circuit is doubled while the voltage stays the same?

Problem 2

If the voltage in a circuit is halved and the resistance remains constant, how does the current change?

Problem 3

A resistor of 10Ω is connected to a 20V battery. Find the current flowing through the resistor.

Problem 4

A current of 0.5A flows through a resistor of 8Ω . Find the voltage across the resistor.

Current and Charge

Problem 5

If a current of 2A flows for 3s, how much charge passes through the circuit?

Problem 6

A battery produces a current of 0.2A for 10s. Calculate the total charge that flows.

Problem 7

How long will it take for 5C of charge to pass through a wire carrying a current of 0.5A?

Electrons and Current Direction

Problem 8

If 0.01C of charge flows through a circuit, how many electrons have passed through?

Problem 9

A current of 3A flows for 4s. Calculate the total number of electrons that moved.

Problem 10

In a conventional current, current is said to flow from positive to negative. Which way do the electrons actually move?

Application Problem

Problem 11

A circuit consists of a 12V battery and a 6Ω resistor.

- a) Calculate the current in the circuit.
- b) How much charge passes through the resistor in 5s?
- c) How many electrons pass through the resistor in that time?
- d) Indicate the direction of electron flow relative to the conventional current.